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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

Daily Teaching : 2 Hours

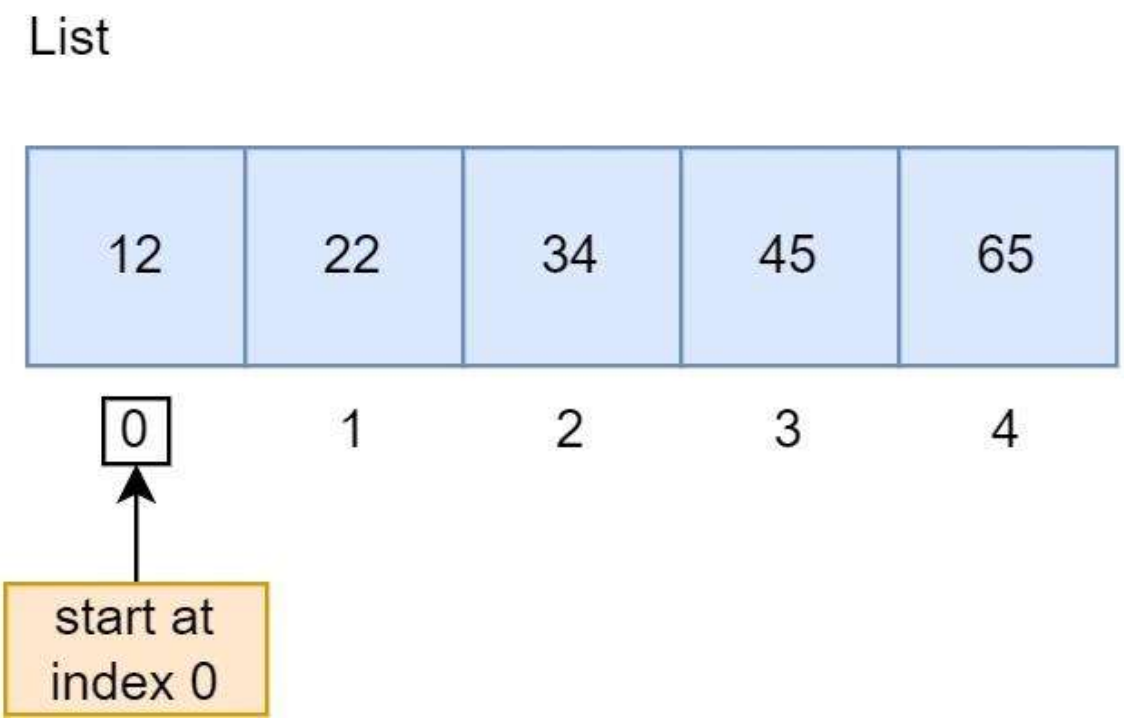
Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 27: Python Lists

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What is a List?





Learning Objectives

After completing this lesson, students will be able to:

- Understand what a List is.
- Create Lists in Python.
- Access List elements.
- Modify List values.
- Add and remove items.
- Use common List methods.
- Write simple List programs.

What is a List?

A **List** is a collection of items stored together.

Think of it like a **school bag**.

Inside your school bag you have:

- Books
- Pens

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- Pencil
- Eraser
- Scale

All these items are inside one bag.

Similarly,

A **List** stores many values inside one variable.

Example

school_bag = ["Book", "Pen", "Pencil", "Eraser"]

Real Life Examples

Example	List
Fruits	Apple, Mango, Banana
Colors	Red, Green, Blue
Students	Rahul, Sneha, Ali
Marks	85, 90, 75
Animals	Lion, Tiger, Cat

Python List Syntax

list_name = [item1, item2, item3]

Example

fruits = ["Apple", "Mango", "Banana"]

Output

['Apple', 'Mango', 'Banana']

Features of Lists

- Ordered

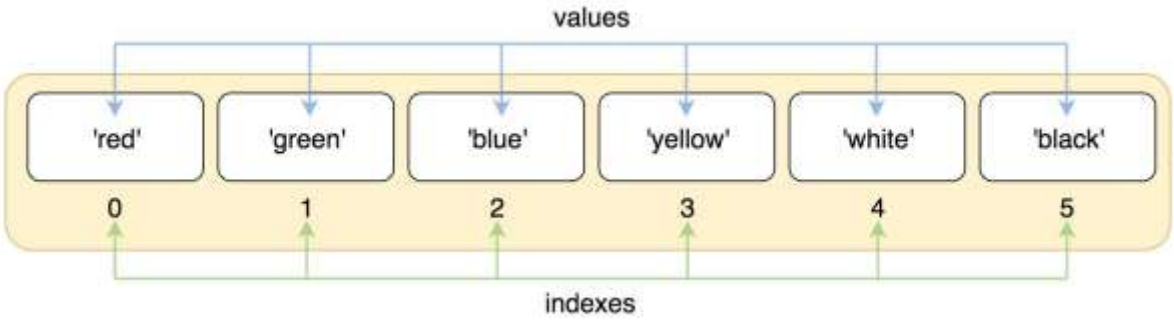
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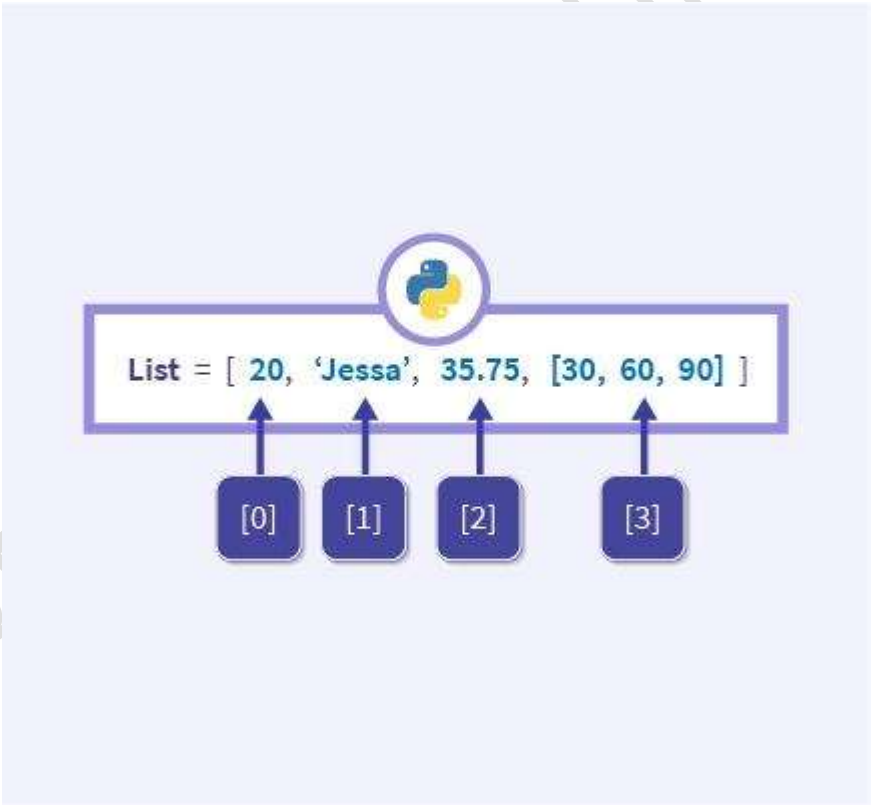
- Changeable
- Allows duplicate values
- Can store different data types

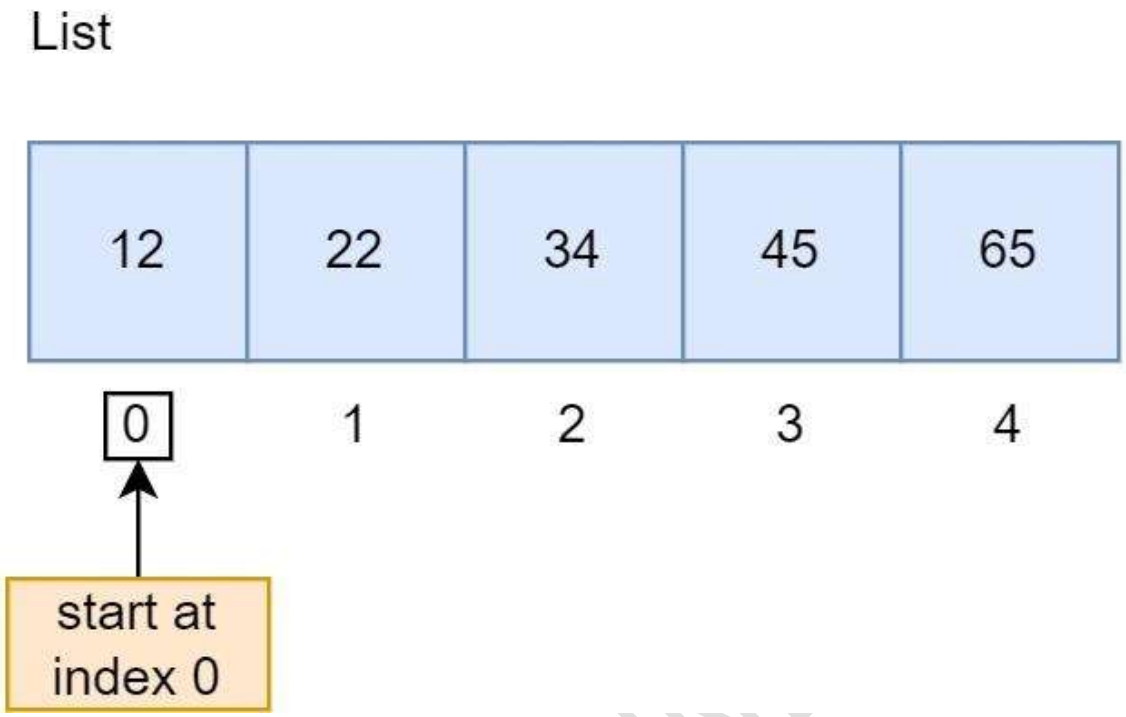
Example

data = ["Ali", 15, 85.5, True]

Interactive Image – List Structure







Creating a List

Example

```
colors = ["Red", "Green", "Blue"]


print(colors)
```

Output

```
['Red', 'Green', 'Blue']
```

Accessing List Items

Lists use **Index Numbers**.

Example

```
fruits = ["Apple", "Mango", "Banana"]


print(fruits[0])
```

Output

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Apple

Second Item

print(fruits[1])

Output

Mango

Third Item

print(fruits[2])

Output

Banana

Negative Index

fruits = ["Apple","Mango","Banana"]

print(fruits[-1])

Output

Banana

Changing List Items

colors = ["Red","Green","Blue"]

colors[1] = "Yellow"

print(colors)

Output

['Red', 'Yellow', 'Blue']

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List Length

```
fruits = ["Apple","Mango","Banana"]


print(len(fruits))
```

Output

3

Adding Items

append()

```
fruits = ["Apple","Mango"]


fruits.append("Banana")


print(fruits)
```

Output

['Apple', 'Mango', 'Banana']

insert()

```
fruits = ["Apple","Banana"]


fruits.insert(1,"Mango")


print(fruits)
```

Output

['Apple', 'Mango', 'Banana']

Removing Items

remove()

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```
fruits = ["Apple","Mango","Banana"]

fruits.remove("Mango")

print(fruits)
```

Output

```
['Apple', 'Banana']
```

pop()

```
fruits = ["Apple","Mango","Banana"]

fruits.pop()

print(fruits)
```

Output

```
['Apple', 'Mango']
```

clear()

```
fruits = ["Apple","Mango"]

fruits.clear()

print(fruits)
```

Output

```
[]
```

Loop Through List

```
fruits = ["Apple","Mango","Banana"]
```

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```
for fruit in fruits:
   print(fruit)
```

Output

Apple

Mango

Banana

Check Item Exists

```
fruits = ["Apple","Banana"]

if "Apple" in fruits:
   print("Found")
```

Output

Found

Sorting Lists

```
numbers = [5,3,8,1]

numbers.sort()

print(numbers)
```

Output

[1,3,5,8]

Reverse List

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```
numbers = [1,2,3,4]


numbers.reverse()


print(numbers)
```

Output

[4,3,2,1]

Copy List

```
list1 = ["Apple","Mango"]


list2 = list1.copy()


print(list2)
```

Output

['Apple', 'Mango']

List Methods

Method	Purpose
append()	Add item
insert()	Add at position
remove()	Remove item
pop()	Remove last item
clear()	Empty list
sort()	Sort list
reverse()	Reverse order

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Method	Purpose
copy()	Copy list
count()	Count occurrences
index()	Find position

Practical Session

Practical 1

Create a list of fruits.

```
fruits = ["Apple","Mango","Banana"]


print(fruits)
```

Practical 2

Add a fruit.

```
fruits.append("Orange")


print(fruits)
```

Practical 3

Remove a fruit.

```
fruits.remove("Apple")


print(fruits)
```

Practical 4

Print every student name.

```
students = ["Ali","Sara","John"]
```

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```
for student in students:
    print(student)
```

Practical 5

Sort marks.

```
marks = [78,95,60,88]

marks.sort()

print(marks)
```

Practical 6

Reverse numbers.

```
numbers = [10,20,30,40]

numbers.reverse()

print(numbers)
```

Practical 7

Find total items.

```
games = ["Chess","Cricket","Football"]

print(len(games))
```

Classroom Activity

Create a list of your:

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- Favorite Fruits
- Favorite Subjects
- Family Members
- Colors

Print all the items.

Fun Activity

My School Bag List

Create a list containing:

Book

Pen

Pencil

Scale

Eraser

Bottle

Print each item one by one using a loop.

Mini Project

Student Marks

marks = [85,90,78,88,95]

print("Student Marks")

for mark in marks:

    print(mark)

print("Total Subjects:", len(marks))

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Common Errors

❌ Missing square brackets

fruits = "Apple","Mango"

✓ Correct

fruits = ["Apple","Mango"]

❌ Wrong Index

print(fruits[5])

If the list has only 3 items, this gives an **IndexError**.

Tips

- Lists use square brackets [ ]
- Index starts from **0**
- Lists are changeable
- Store many values together

Summary

- A List stores multiple items.
- Lists are ordered and changeable.
- Access items using indexes.
- Add items with append() and insert().
- Remove items with remove() and pop().
- Sort with sort().
- Reverse with reverse().
- Use for loops to display items.

Quiz

Choose the correct answer.

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1. Which symbol is used to create a List?

A) { }

B) ( )

C) [ ]

D) < >

✅ **Answer:** C

2. What is the first index of a List?

A) 1

B) 2

C) 0

D) -1

✅ **Answer:** C

3. Which method adds an item to the end of a List?

A) remove()

B) append()

C) pop()

D) clear()

✅ **Answer:** B

4. Which method sorts a List?

A) sort()

B) reverse()

C) clear()

D) insert()

✅ **Answer:** A

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5. Which function gives the number of items in a List?

A) size()

B) length()

C) len()

D) count()

✅ **Answer:** C

Home Assignment

1. Create a list of **5 animals** and print them.
2. Add one more animal using append().
3. Remove one animal using remove().
4. Print the **first** and **last** animal.
5. Sort the list alphabetically.
6. Reverse the list.
7. Create a list of **5 subject marks** and print the highest mark after sorting.

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